Plan — from here to a fabricated board

Current state: `STATE.md`. Superseded plans: `docs/archive/`.

> **Update 2026-09-23:** the decision below was made — the fabricated board is the cascade N=3 with the end cavities trimmed −0.3 mm (`l_cav1` = 33.939 mm). Verdict, JLCPCB order notes and the fab package: `final_check/REPORT.md`, `final_check/fab/`.

The design work is **done**, in the sense that the design space has been mapped and the remaining gap is understood rather than mysterious. What is left is a decision, then fabrication outputs.

The decision that gates everything

The specification cannot be met by this topology. Bandwidth and return loss are in direct opposition on every geometry parameter (see `STATE.md`, "the central finding"), and either can be met alone:

option	bandwidth	return loss	note
good match (current design)	316.6 MHz	**−10.76 dB**	2.2× the target band
good bandwidth	**145.6 MHz**	−1.75 dB	essentially unmatched
spec	143.5 MHz	−16 dB	not reachable together here

Decision taken 2026-09-03: the good-match end. The current design is that choice, optimised and confirmed.

Three ways forward, in the order I would consider them:

1. **Accept and fabricate.** Write the deviation down explicitly — a 316.6 MHz, −10.76 dB filter at 2.90 GHz — and order the board. Reasonable if the real requirement is "a working S-band cavity filter" rather than the literal numbers.
2. **Change the topology.** N=2 dual-mode has one internal coupling and one external coupling; that is not enough freedom to place two poles *and* match them. A cascade of two single-mode cavities has an extra degree of freedom, at roughly 1.5× the board length. The parked cascade N=3 data is in `golden/cascade_N3/`.
3. **Relax the specification.** If 5 % FBW is a nice-to-have and return loss is the real requirement, the current design already satisfies the requirement that matters.

This is a user decision, not a technical one. Nothing further should be optimised until it is made.

> **Decision revised, 2026-09-04: option 2, cascade N=3.** Closer to spec on every axis (−13.37 dB / 189 MHz, fine-confirm −14.92 dB / 188 MHz, vs dual-mode's −10.76 dB / 316.6 MHz), at ~1.5× the board length (166.4 vs 119.0 mm). Steps 2 and 3 below are now DONE for this design — see the status block that follows — not merely planned.
>
> **Caveat added 2026-09-06, found during the CST/Palace audit (`docs/audit_cst_palace/`): the −13.37 dB S11 above is NOT mesh-converged.** `output/data/mesh_convergence_cascade_N3.json` (already in the repo, predates this audit) refines the mesh from (3.5, 0.6, 6.0) mm to (2.5, 0.45, 6.0) mm and S11w gets **4 dB worse** (−13.37 → −9.34 dB), while fc/BW/IL barely move. A separate 61-point sweep with refined ports (`docs/audit_cst_palace/AUDITORIA.md` §4) independently lands in the same place: worst in-band S11 −10.56 dB. So cascade N=3's real match is probably close to dual-mode's −10.76 dB, not the −13.37 dB this decision was made on — the "closer to spec on every axis" claim above needs re-checking against a real mesh-convergence study before it's trusted.

Status, 2026-09-04 (cascade N=3, the board actually being ordered)

- PEC regression (121-point sweep, after today's real-copper port rework and a board-outline fix): reproduces the golden to every digit — fc 2.879 GHz, IL −1.85 dB, BW 189 MHz, S11w −13.37 dB.
- Real copper measured (exterior ground planes only, Palace's `Boundaries.Conductivity` is exterior-only so via barrels and the trace stay PEC): IL −1.85 → −2.44 dB, S11w −14.92 → −13.82 dB, shape and bandwidth unmoved. Recorded in `golden/cascade_N3/...json`.
- CST `.bas` (`siw cst`) and KiCad board + gerbers/drill/STEP/SVG (`siw kicad`) both generate from the same `Geometry`/launch objects the mesher uses, not a second description of the board.
- `siw kicad --check` runs 29 file-level + connector-fit checks against the actual generated gerbers/drill/`.bas` (not the code that wrote them) and writes `FAB_CHECKLIST.md`. All 29 pass on the current geometry.
- Taoglas EMPCB.SMAFSTJ.B.HT verified against its real datasheet (`docs/sma_connector/`): pin and ground-leg-pitch numbers were already exact; the finished board (0.795 mm) fits the connector's 0.8 mm/0.79 mm limits with no margin, which is why 0.5 oz copper is a requirement, not a preference.

- **Still open**: the deviation from spec itself (table above) is accepted, not fixed, and remains what it is — write it into the fab order. No mesh convergence study exists for `fc` (see "Later" below — never done, would put an error bar on every frequency quoted here, including the +0.31 % currently recorded against spec).

Step 1 — Optional: re-polish on real copper (dual-mode only; moot for cascade)

This was scoped for the dual-mode design before the decision was revised. Cascade N=3's real-copper number is already measured (see the status block above) at zero extra search cost, so this step does not apply to the board actually being built. Left here in case dual-mode is ever revisited:

```
siw design --topology dualmode --N 2 \
    --seed a=57.158,w_iris=15.304,x_iris=28.685,y_wall=9.849,w_ext=16.352 \
    --span 2 --patience 14 --evals 12 --w-bw 0 --real-metal --device CPU --np 4
```

Real copper shifts that design by 0.32 dB of insertion loss and 0.21 dB of return loss without changing its shape, close to the coarse-mesh noise floor; twelve evaluations cost about 6.5 hours. Do **not** pass `--enig`: it implies ENIG + soldermask + silkscreen, all unresolvable at this mesh density and destroy the response.

Step 2 — Fabrication outputs — DONE for cascade N=3

```
siw kicad --topology cascade --N 3 \
    --params l_cav1=33.939,l_cav2=35.503,w_iris=14.416,w_ext=13.96,l_taper=15.738,w_ms=1.663
    --launch cpwg
```

- ☒ CST `.bas` export (`siw cst`) — StoreParameter'd, real Taoglas SMA coax stubs + coaxial ports. Visual check in CST is still the human's to do; the generator-side unit bugs found while checking it are fixed.
- ☒ Gerbers with the soldermask **fully open** (the decision made, not opened only over the feed as this step used to say — see `FAB_NOTES.md`'s explanation of how "no mask" is expressed in gerber).
- ☒ JLCPCB DRC re-check against `config.py`'s `FAB_RULES`, plus five connector-fit checks against the real Taoglas datasheet — `siw kicad --check`, 29/29 passing, `FAB_CHECKLIST.md`.
- ☒ ENIG finish, vias open/unfilled — stated in `FAB_NOTES.md`.

Step 3 — Order the board

Only once the decision above is written down. Do not ship a board that silently misses spec. For cascade N=3 the deviation is on record in `FAB_CHECKLIST.md`'s "Known deviations from specification" table, sourced from `golden/cascade_N3/`.

Later

- **Paper** — "AI-Driven Design for SIW Filters". The strongest material is already collected, and most of it is negative or methodological:
  - a *structural* obstacle to black-box optimisation — an objective term that is discontinuous in the external coupling, plus the discovery that the objective's own bandwidth weight steered every search away from the best design;
  - **fifteen silent modelling bugs**, none of which raised an error;
  - the calibrate-and-invert failure with its mechanism (a dual-mode cavity's two resonators are two modes of one cavity, so no proxy can isolate one);
  - the bandwidth/match opposition as a measured property of the topology;
  - the limits of thin-layer modelling at practical mesh densities.
- **Mesh convergence study** for `fc` — never done; would put an error bar on every frequency in this project.

Do not redo

- **The fabrication stack study.** Soldermask, silkscreen and ENIG are unresolvable at this mesh density; the measured "penalties" scale with layer *thinness*, which is the signature of sliver elements, not physics. Rerunning them produces the same meaningless numbers.
- **Volumetric lossy-conductor vias.** Same problem, and the hollow-barrel geometry is gated on them, which is why tented-vs-open is unanswerable.
- **Calibrate-and-invert on dual-mode.** The Qe proxy measures a different structure (3.01 GHz vs the design's 2.89 GHz); proxy loaded-Q 8.29 against the assembled filter's 8.66 proves it. Two coupled resonators cannot share the loaded Q of one.
- **Compact N=4 dual-mode cascade** — five mechanisms plus a joint search all hit the same ceiling. Raw data in `docs/geometry_exploration/`.
- **Lossy-conductor boundary conditions on internal surfaces** — Palace's `Conductivity` BC is exterior-only. It works correctly on the exterior ground planes, which is why the real-copper result is trustworthy.

- **A second launch definition.** There is one, in `topologies/launch.py`, plus an explicitly-named pad-only alternative. Forking it caused three separate bugs.